File Attributes

1 Basic Idea

Definition

File attributes are extra pieces of information stored by the operating system about a file, in addition to the file name and file data.

- Every file has a name.
- Every file has data.
- Operating systems also store extra information about each file.
- Examples include file size, owner, protection, and modification time.
- These extra items are also called **metadata**.
- The exact set of attributes varies from system to system.

Core Point

Attributes describe and control a file; they are data about the file, not usually the file's user-visible contents.

2 Common File Attributes

Attribute	Meaning
Protection	Who can access the file and in what way.
Password	Password needed to access the file.
Creator	ID of the person who created the file.
Owner	Current owner of the file.
Read-only flag	0 for read/write; 1 for read only.
Hidden flag	0 for normal; 1 for do not display in listings.
System flag	0 for normal file; 1 for system file.
Archive flag	0 for backed up; 1 for needs backup.
ASCII/binary flag	0 for ASCII file; 1 for binary file.
Random access flag	0 for sequential access only; 1 for random access.
Temporary flag	0 for normal; 1 for delete on process exit.
Lock flags	0 for unlocked; nonzero for locked.
Record length	Number of bytes in a record.
Key position	Offset of the key within each record.
Key length	Number of bytes in the key field.
Creation time	Date and time the file was created.
Time of last access	Date and time the file was last accessed.
Time of last change	Date and time the file was last changed.
Current size	Number of bytes currently in the file.
Maximum size	Maximum number of bytes the file may grow to.

3 Protection Attributes

Protection

Protection attributes say who may access a file and what operations they may perform.

- Protection controls access to a file.
- It may specify read, write, execute, append, or delete permissions.
- The creator attribute records who created the file.
- The owner attribute records who currently owns the file.
- Some systems require a password to access a file.
- In those systems, the password is stored as part of the file metadata.

Security Note

Real systems should not store plain-text passwords directly as file attributes. The textbook idea is about access metadata, not a recommended password-storage design.

4 Flag Attributes

Flags

Flags are bits or short fields that enable or describe special file properties.

Flag	Use
Read-only	Prevents writing to the file.
Hidden	Prevents the file from appearing in ordinary file listings.
System	Marks the file as part of the operating system.
Archive	Records whether the file needs backup.
ASCII/binary	Records whether the file is text or binary.
Random access	Records whether random access is supported.
Temporary	Marks the file for deletion when the creating process exits.
Lock	Shows that the file or part of it is locked.

5 Archive Flag

- The archive flag helps backup software.
- A backup program clears the archive flag after backing up a file.
- The operating system sets the archive flag when the file is changed.
- Later, the backup program can check the flag.
- Files with the flag set need to be backed up again.

Backup Use

The archive flag is a small metadata bit that helps backup programs identify changed files.

6 Temporary and Hidden Files

- A hidden file does not appear in normal file listings.
- Hidden files are often used for configuration or system-related data.
- A temporary file may be deleted automatically.
- The temporary flag can tell the OS to delete the file when the creating process terminates.
- These flags affect how the system manages or presents the file.

7 Record and Key Attributes

Record-Key Metadata

Record length, key position, and key length are useful for file systems that support record lookup by key.

- These fields are not needed for ordinary byte-sequence files.
- Record length says how many bytes are in each record.
- Key position says where the key begins inside a record.
- Key length says how many bytes the key field occupies.
- These attributes help the OS find records by key.

8 Time Attributes

Time attribute	Meaning
Creation time	When the file was created.
Time of last access	When the file was most recently read or otherwise accessed.
Time of last change	When the file was most recently modified.

- Time attributes are useful for users, tools, and the OS.
- A compiler can compare source-file time and object-file time.
- If the source file is newer than the object file, recompilation is needed.
- Backup tools can use timestamps to identify changed files.
- File managers can sort files by creation, access, or modification time.

9 Size Attributes

File Size

The current size records how many bytes are currently in the file.

- Current size tells how large the file is now.
- Maximum size tells how large the file is allowed to become.
- Some older mainframe systems required maximum size at file creation time.
- That allowed the OS to reserve storage in advance.
- Modern workstation and personal-computer operating systems usually do not require this.

10 Grouped View

Group	Examples
Protection	Protection, password, creator, owner.
Flags	Read-only, hidden, system, archive, temporary, lock.
Record/key data	Record length, key position, key length.
Times	Creation time, last access time, last change time.
Sizes	Current size, maximum size.

Final Point

File attributes are metadata used to protect, classify, locate, manage, back up, and interpret files.